Blinding Effectiveness

Libraries and Functions

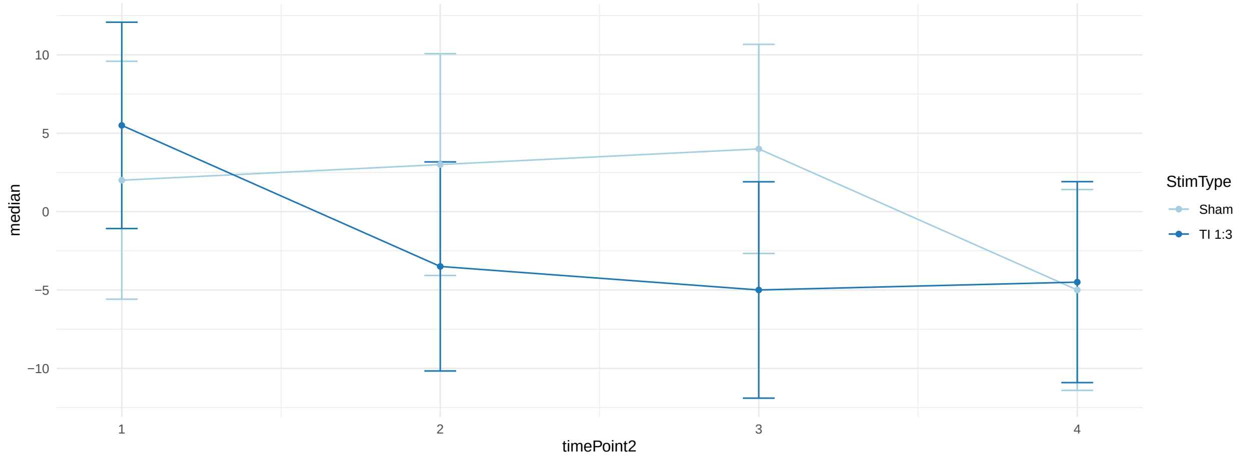
```
# Clear envirnonment
rm(list = ls())

# Get Libraries and Functions
source("~/Dropbox (Personal)/TI_paper/ti-paper/Data/Functions.R")
```

Import data and add labels

```
# Import data
setwd("~/Dropbox (Personal)/TI_paper/ti-paper/Data/BehaviourStudy/Blinding")
data_raw <- read_csv("StimGuessDataLong.csv")

# Assign appropriate Stim factors
data_raw$S_ID <- data_raw$SubjectNumber
data_raw$StimType <- data_raw$Stimulation
data_raw$StimType <- factor(data_raw$StimType)
levels(data_raw$StimType) <- c("Sham", "TI 1:3")

# Assign factors
data_raw$StimType <- factor(data_raw$StimType)
data_raw$S_ID <- factor(data_raw$S_ID)
data_raw$Session <- factor(data_raw$Session)
```

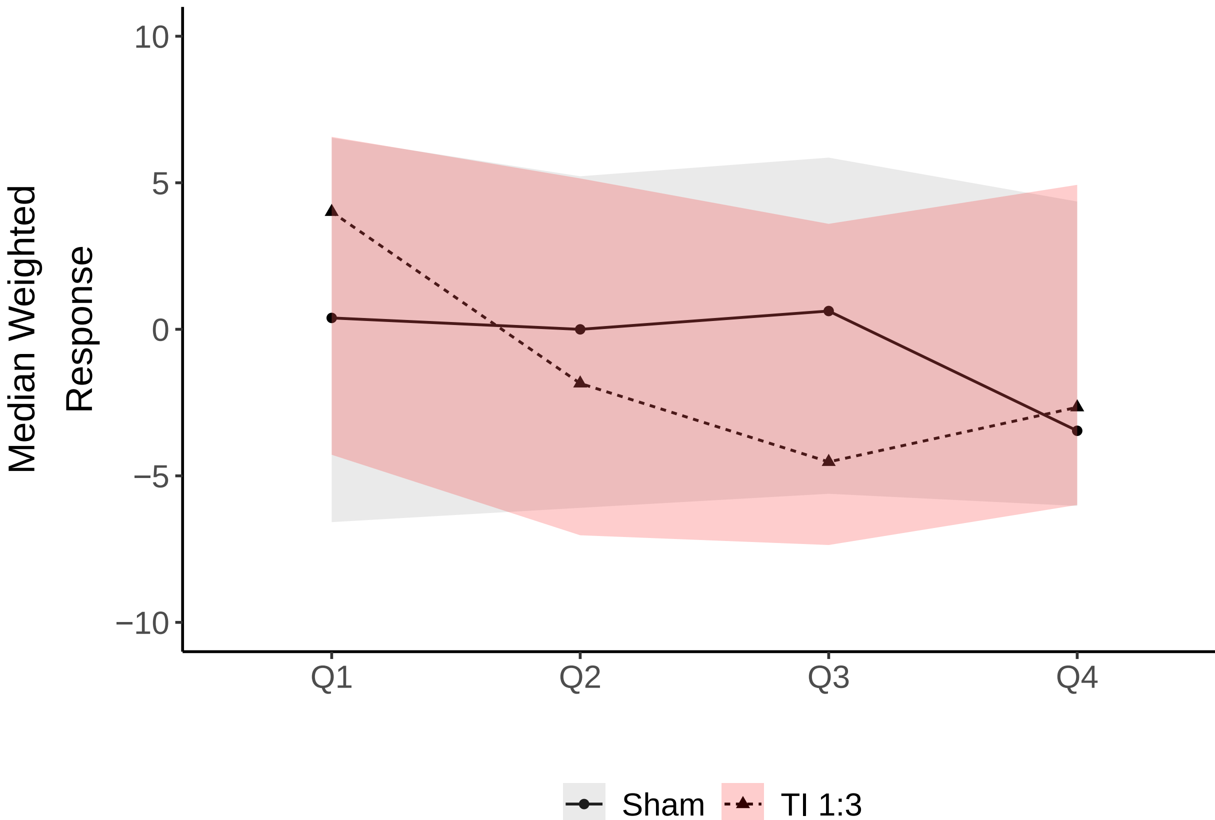
Plot data

```
g_data <- ddply(data_raw, c("StimType","timePoint2"), summarise,
         mean = mean(WeightedScore),
         median = median(WeightedScore),
         sd = sd(WeightedScore),
         se = se(WeightedScore)
)

ggplot(g_data, aes(x=timePoint2, y=median, group=StimType, color=StimType)) +
    geom_errorbar(aes(ymin=median-sd, ymax=median+sd), width=.1) +
    geom_line() + geom_point()+
   scale_color_brewer(palette="Paired")+theme_minimal()
```




```
bdata <- data_raw

# Create matrix split by StimType and time Point
sp <- split(bdata, list(bdata$StimType, bdata$timePoint2))

# Set seed for reproducibility
set.seed(1)

# Define Functions
boot.median = function(data,index){
  d = data[index,]
  return(median(d$WeightedScore))
}


# kernelboot package to calculate median and 95 CI
# this uses a smooth bootstrap technique
library(kernelboot)

ysb <- lapply(sp, function(x){
  #browser()
  # Median - summary gives the mean of the median
  bootobject.median.g <- kernelboot(x, boot.median, R = 10000, kernel = "gaussian", bw =1)
  median_gau <- median(bootobject.median.g$boot.samples)

  s_gau = bootobject.median.g$boot.samples

  data.frame(median_gau, summary(bootobject.median.g))

})

## Plot with different kernels
smboot <- do.call(rbind, ysb)
smboot$StimType <- c("Sham", "TI 1:3", "Sham", "TI 1:3", "Sham", "TI 1:3", "Sham", "TI 1:3")
smboot$time <- c("1", "1", "2", "2", "3", "3", "4", "4")
```

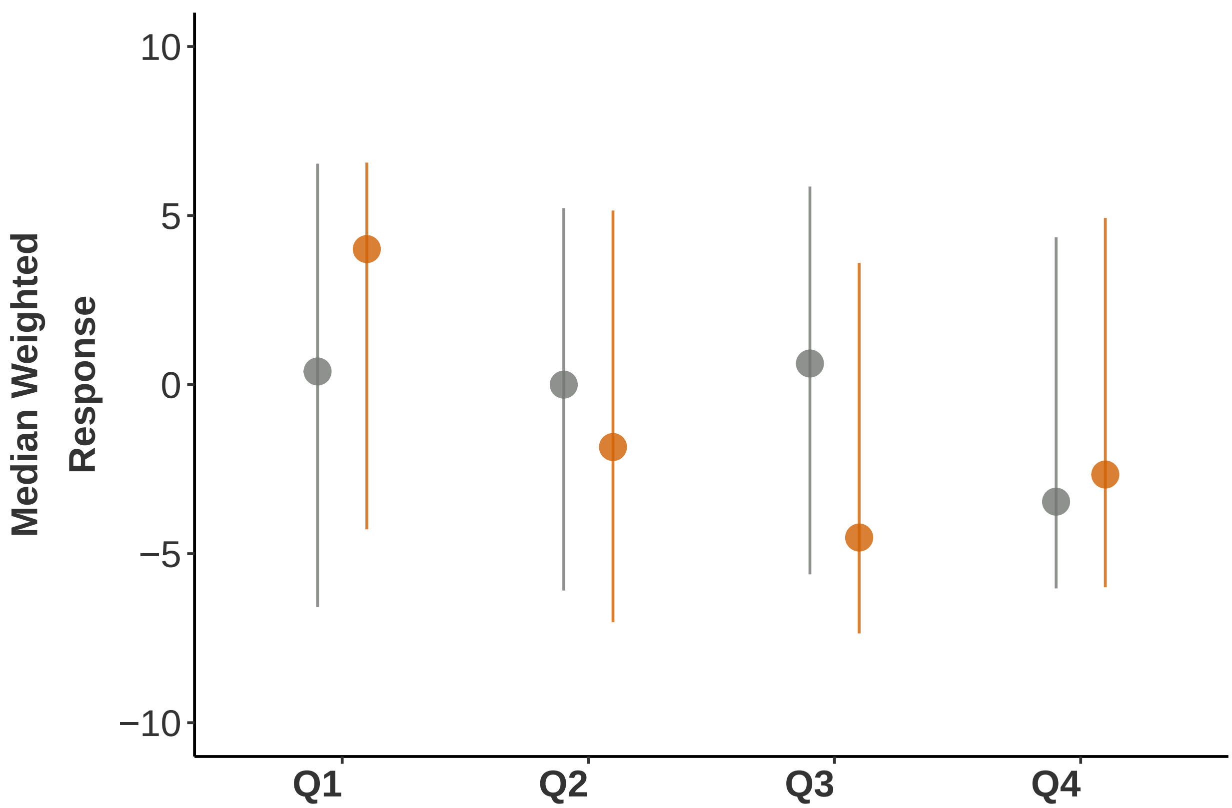
```
p.median.gau <- ggplot(smboot, aes(x=time, y=`mean`, group=StimType)) +
  geom_line(aes(linetype=StimType))+
  geom_point(aes(shape=StimType)) +
  geom_ribbon(aes(time, ymin = `X2.5.`, ymax = `X97.5.`, fill = StimType), alpha = .2) +
  ylim(-10, 10) +  theme_classic() + scale_color_manual(values = c("#999999", "##fc0707")) +
  scale_fill_manual(values = c("#999999","#fc0707")) +
  xlab("") + ylab(expression(atop("Median Weighted", paste("Response")))) + theme(legend.title = elemen
                                                                   axis.title = element_text(size = 14),
                                                                   axis.text = element_text(size = 12), legend

p.median.gau
```



Bootstrapping

```
## Plot Gaussian different style
p.median.gau2 <- ggplot(data = smboot, mapping=aes(x=time, y=`mean`, ymin=`X2.5.`, ymax = `X97.5.`, grou
  geom_pointrange(size=0.9, alpha=0.8, position = position_dodge(width = 0.4),
                color= c("#747674", "#D16103", "#747674", "#D16103", "#747674", "#D16103", "#747674",
  ylim(-10, 10) +
  theme(plot.title = element_text(color="black", size=14, face="bold", hjust = 0.5),
        axis.text.x = element_text(color = "grey20", size = 14, angle = 0, face = "bold", hjust = 1),
        axis.text.y = element_text(color = "grey20", size = 14, angle = 0, face = "plain"),
        axis.title.x = element_text(color = "grey20", size = 14, angle = 0, face = "bold"),
        axis.title.y = element_text(color = "grey20", size = 14, angle = 90, face = "bold"),
        legend.text = element_blank(),
        legend.background = element_blank(),
        legend.key=element_blank(),
```

```
        panel.background = element_blank(),
        legend.title=element_blank()) +
  theme(panel.grid.major = element_blank(),
        panel.grid.minor = element_blank(), axis.line = element_line(colour = "black")) +
  theme(strip.text.x = element_text(size = 14),
        strip.text.y = element_text(size = 20)) +
   theme(legend.title = element_blank(), legend.position="bottom",
                                                                 axis.title = element_text(size = 14),
                                                                 axis.text = element_text(size = 12), legend

p.median.gau2
```




```
ggsave(p.median.gau2, filename =  "Figure_5G.pdf", width = 9, height = 7, dpi = 300, units = "cm", devi
```

```
bdata <- data_raw
xtabs( ~ WeightedScore + StimType + timePoint2, bdata)
```

LME

```
## , , timePoint2 = 1
##
```

```
##              StimType
## WeightedScore Sham TI 1:3
##           -10    1      1
##           -9     1      0
##           -8     3      1
##           -7     1      1
##           -6     3      2
##           -5     0      1
##           -4     1      1
##           -3     0      0
##           2      1      0
##           3      0      2
##           4      1      0
##           5      0      1
##           6      3      4
##           7      2      3
##           8      0      2
##           9      1      0
##           10     3      1
##
## , , timePoint2 = 2
##
##              StimType
## WeightedScore Sham TI 1:3
##           -10    2      1
##           -9     1      0
##           -8     0      4
##           -7     2      3
##           -6     3      1
##           -5     1      0
##           -4     1      1
##           -3     0      1
##           2      0      0
##           3      2      0
##           4      2      1
##           5      1      3
##           6      1      2
##           7      2      3
##           8      1      0
##           9      0      0
##           10     2      0
##
## , , timePoint2 = 3
##
##              StimType
## WeightedScore Sham TI 1:3
##           -10    1      1
##           -9     0      1
##           -8     1      4
##           -7     2      3
##           -6     3      0
##           -5     2      2
##           -4     1      2
##           -3     0      0
```

```
##           2      0      0
##           3      0      1
##           4      3      1
##           5      1      0
##           6      1      1
##           7      5      1
##           8      0      1
##           9      0      2
##           10     1      0
##
## , , timePoint2 = 4
##
##              StimType
## WeightedScore Sham TI 1:3
##           -10    1      1
##           -9     1      1
##           -8     1      2
##           -7     0      1
##           -6     6      1
##           -5     2      4
##           -4     1      1
##           -3     1      1
##           2      0      0
##           3      1      0
##           4      0      1
##           5      3      2
##           6      0      1
##           7      3      4
##           8      0      0
##           9      0      0
##           10     1      0
```

```r
# Not balanced - subject 1 only had measurements from Sham

m0 <- lmer(WeightedScore ~ StimType*factor(timePoint2) + (1|S_ID), data=bdata)
m1 <- lmer(WeightedScore ~ StimType*factor(timePoint2) + (1|S_ID) + (1|Session), control=lmerControl(op
```

```
## boundary (singular) fit: see help('isSingular')
```

```r
anova(m0, m1)
```

```
## refitting model(s) with ML (instead of REML)
```

```
## Data: bdata
## Models:
## m0: WeightedScore ~ StimType * factor(timePoint2) + (1 | S_ID)
## m1: WeightedScore ~ StimType * factor(timePoint2) + (1 | S_ID) + (1 | Session)
##    npar    AIC    BIC  logLik deviance Chisq Df Pr(>Chisq)
## m0   10 1065.4 1096.4 -522.71   1045.4
## m1   11 1067.4 1101.5 -522.71   1045.4     0  1          1
```

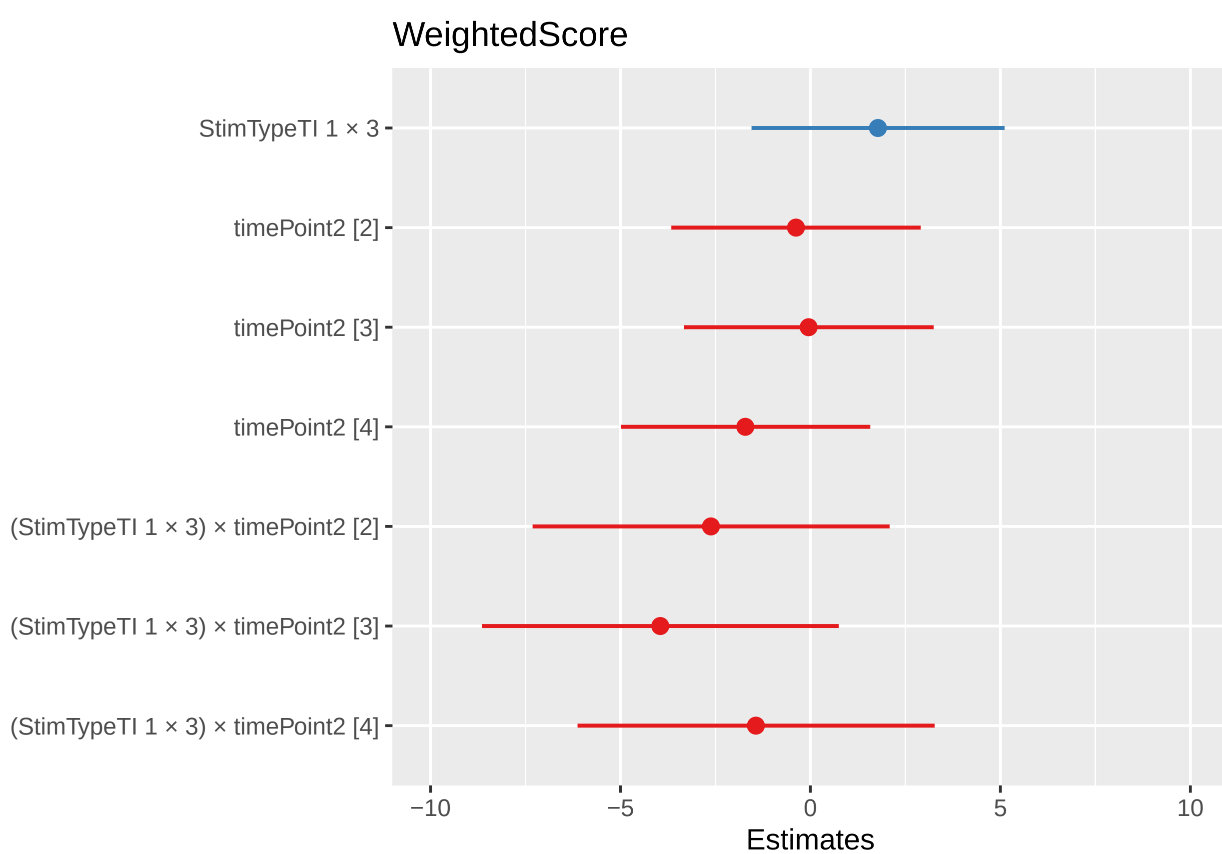
```r
lmtest:::lrtest(m0, m1)
```

```
## Likelihood ratio test
##
## Model 1: WeightedScore ~ StimType * factor(timePoint2) + (1 | S_ID)
## Model 2: WeightedScore ~ StimType * factor(timePoint2) + (1 | S_ID) +
##     (1 | Session)
##   #Df LogLik Df Chisq Pr(>Chisq)
## 1  10 -513.2
## 2  11 -513.2  1     0          1
```

```r
# keeping m1

Anova(m1)
```

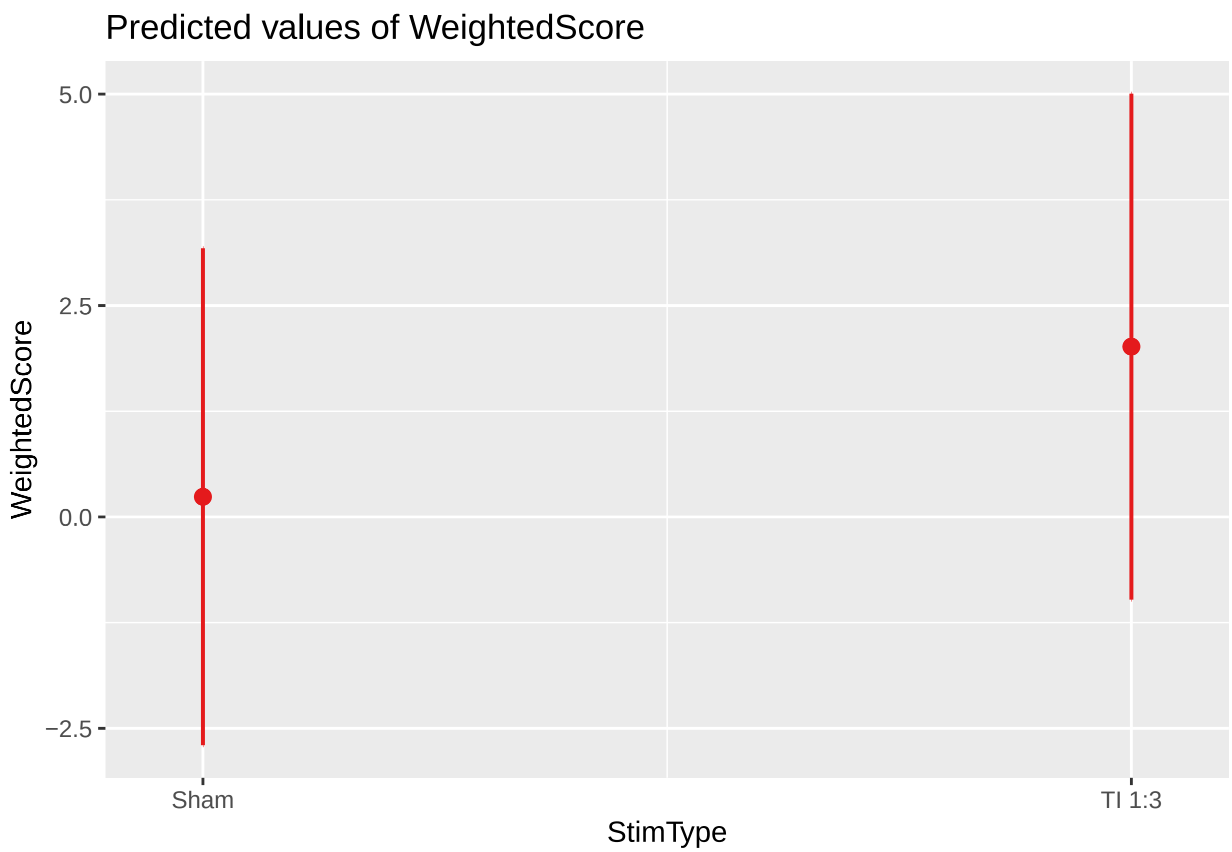
```
## Analysis of Deviance Table (Type II Wald chisquare tests)
##
## Response: WeightedScore
##                              Chisq Df Pr(>Chisq)
## StimType                    0.0710  1     0.7899
## factor(timePoint2)          4.7361  3     0.1922
## StimType:factor(timePoint2) 3.0186  3     0.3888
```

```r
plot_model(m1)
```

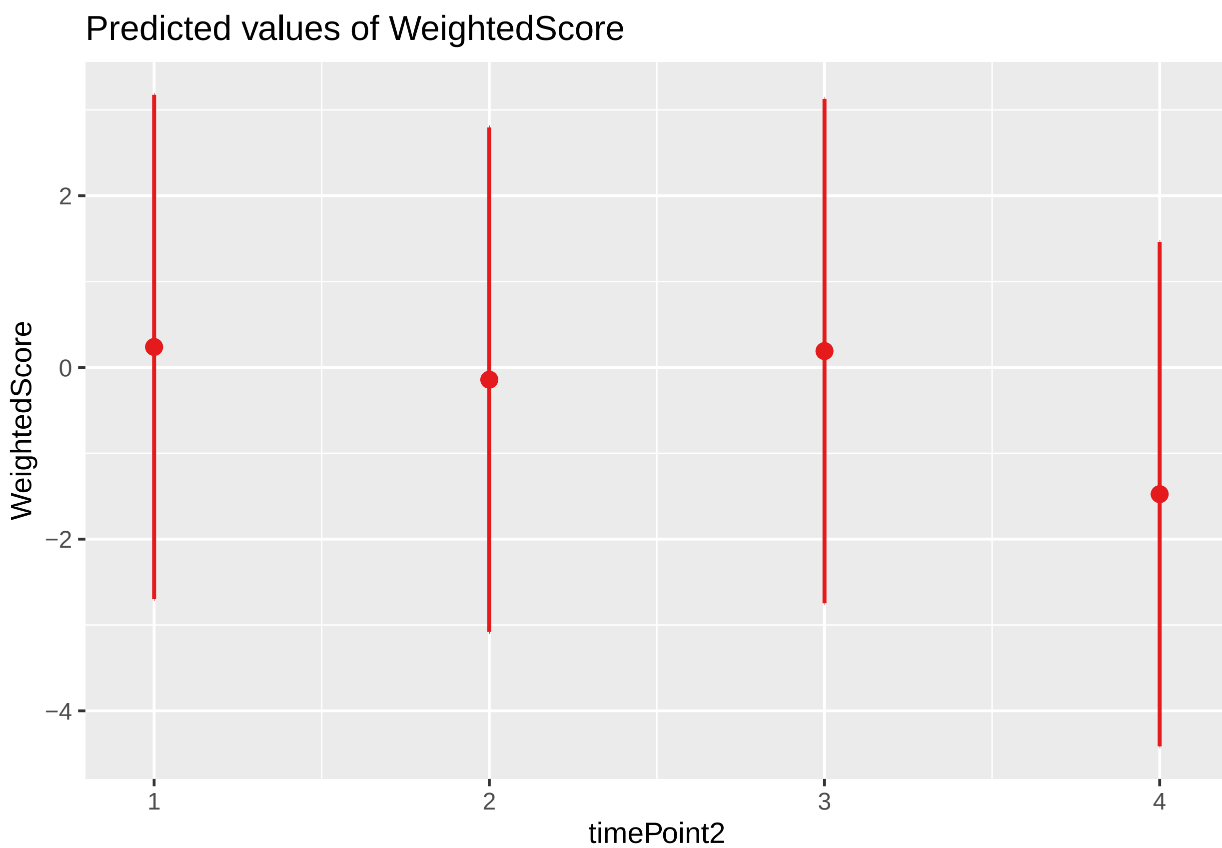


```r
plot_model(m1, type = "pred")
```

```
## $StimType
```




```
##
## $timePoint2
```




```
wilcox.test(subset(bdata, StimType=="Sham" & S_ID!="1")$WeightedScore, subset(bdata, StimType=="TI 1:3"
```

```
##
##  Wilcoxon signed rank test with continuity correction
##
## data:  subset(bdata, StimType == "Sham" & S_ID != "1")$WeightedScore and subset(bdata, StimType == "
## V = 1063.5, p-value = 0.5434
## alternative hypothesis: true location shift is not equal to 0
```

```
bdata$stimGuessRating  <- factor(bdata$stimGuessRating,
                                 ordered=TRUE)

library(ordinal)
```

```
##
## Attaching package: 'ordinal'

## The following objects are masked from 'package:brms':
##
##     ranef, VarCorr

## The following objects are masked from 'package:lme4':
##
##     ranef, VarCorr
```

```
## The following object is masked from 'package:dplyr':
##
##     slice
```

```
library(RVAideMemoire)
```

```
## *** Package RVAideMemoire v 0.9-81-2 ***

##
## Attaching package: 'RVAideMemoire'

## The following object is masked _by_ '.GlobalEnv':
##
##     se

## The following objects are masked from 'package:ppcor':
##
##     pcor, pcor.test

## The following object is masked from 'package:lme4':
##
##     dummy
```

```
m3 = clmm(stimGuessRating ~ StimType*as.factor(timePoint2) + (1|S_ID), data = bdata, Hess = TRUE)
Anova.clmm (m3)
```

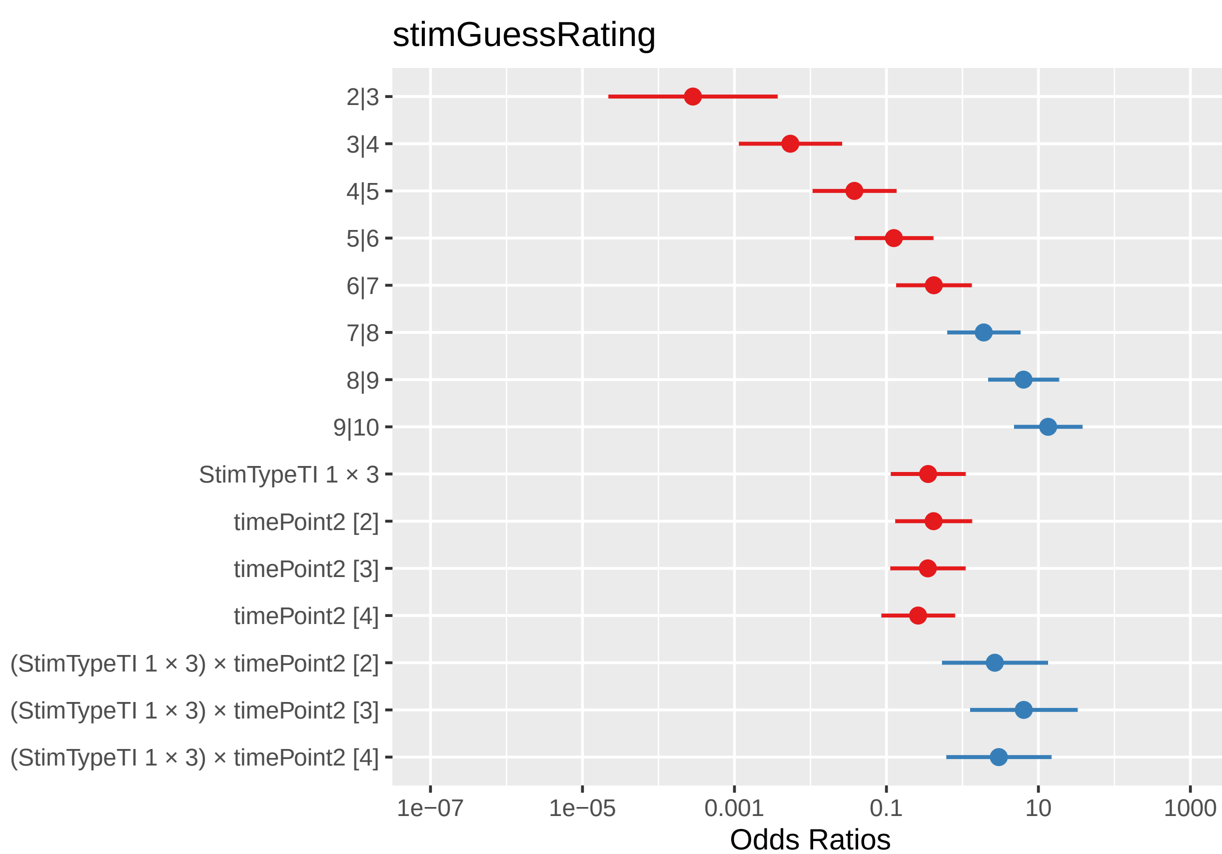
```
## Analysis of Deviance Table (Type II tests)
##
## Response: stimGuessRating
##                                 LR Chisq Df Pr(>Chisq)
## StimType                          0.0458  1     0.8306
## as.factor(timePoint2)             4.2362  3     0.2371
## StimType:as.factor(timePoint2)    5.1746  3     0.1594
```

```
emmeans(m3, specs = pairwise ~ StimType | timePoint2, type = "response")
```

```
## $emmeans
## timePoint2 = 1:
##  StimType emmean    SE  df asymp.LCL asymp.UCL
##  Sham      1.813 0.586 Inf     0.664      2.96
##  TI 1:3    0.774 0.568 Inf    -0.340      1.89
##
## timePoint2 = 2:
##  StimType emmean    SE  df asymp.LCL asymp.UCL
##  Sham      0.939 0.587 Inf    -0.211      2.09
##  TI 1:3    0.881 0.572 Inf    -0.240      2.00
##
## timePoint2 = 3:
##  StimType emmean    SE  df asymp.LCL asymp.UCL
##  Sham      0.765 0.577 Inf    -0.365      1.90
##  TI 1:3    1.585 0.594 Inf     0.421      2.75
```

```
##
## timePoint2 = 4:
##  StimType emmean    SE  df asymp.LCL asymp.UCL
##  Sham      0.470 0.560 Inf    -0.628      1.57
##  TI 1:3    0.534 0.590 Inf    -0.622      1.69
##
## Confidence level used: 0.95
##
## $contrasts
## timePoint2 = 1:
##  contrast      estimate    SE  df z.ratio p.value
##  Sham - TI 1:3   1.0390 0.576 Inf   1.803  0.0714
##
## timePoint2 = 2:
##  contrast      estimate    SE  df z.ratio p.value
##  Sham - TI 1:3   0.0575 0.581 Inf   0.099  0.9211
##
## timePoint2 = 3:
##  contrast      estimate    SE  df z.ratio p.value
##  Sham - TI 1:3  -0.8200 0.589 Inf  -1.392  0.1640
##
## timePoint2 = 4:
##  contrast      estimate    SE  df z.ratio p.value
##  Sham - TI 1:3  -0.0635 0.574 Inf  -0.111  0.9119
```

```
plot_model(m3)
```



```r
setwd("~/Dropbox (Personal)/TI_paper/ti-paper/Data/BehaviourStudy/Blinding")
post_stim <- read_csv("PostStimQ.csv")
```

Post-Stimulation Questionnaire

```
## Rows: 42 Columns: 19
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr  (1): StimType
## dbl (18): S_ID, adv_itching, adv_tingling, adv_skinredness, adv_headache, ad...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
# Assign factors
post_stim$StimType <- factor(post_stim$StimType)
post_stim$S_ID <- factor(post_stim$S_ID)

# Perform Wilcoxon signed-rank tests

adv <- colnames(post_stim)
adv <- adv[-c(1:2)]

library(coin)
library(rstatix)
```

```
##
## Attaching package: 'rstatix'
##
## The following objects are masked from 'package:coin':
##
##     chisq_test, friedman_test, kruskal_test, sign_test, wilcox_test
##
## The following object is masked from 'package:MASS':
##
##     select
##
## The following objects are masked from 'package:plyr':
##
##     desc, mutate
##
## The following object is masked from 'package:stats':
##
##     filter
```

```r
lapply(post_stim[,adv], function(x) coin::wilcox_test(x ~ post_stim$StimType, paired = TRUE))
```

```
## Warning in .local(.Object, ...): The conditional covariance matrix has zero
## diagonal elements
```

```
## Warning in .local(.Object, ...): The conditional covariance matrix has zero
## diagonal elements

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## diagonal elements

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## diagonal elements

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## diagonal elements

## Warning in .local(.Object, ...): The conditional covariance matrix has zero
## diagonal elements

## $adv_itching
##
##  Asymptotic Wilcoxon-Mann-Whitney Test
##
## data:  x by post_stim$StimType (Sham, TI 1:3)
## Z = -2.3538, p-value = 0.01858
## alternative hypothesis: true mu is not equal to 0
##
##
## $adv_tingling
##
##  Asymptotic Wilcoxon-Mann-Whitney Test
##
## data:  x by post_stim$StimType (Sham, TI 1:3)
## Z = 0, p-value = 1
## alternative hypothesis: true mu is not equal to 0
##
##
## $adv_skinredness
##
##  Asymptotic Wilcoxon-Mann-Whitney Test
##
## data:  x by post_stim$StimType (Sham, TI 1:3)
## Z = NaN, p-value = NA
## alternative hypothesis: true mu is not equal to 0
##
##
## $adv_headache
##
##  Asymptotic Wilcoxon-Mann-Whitney Test
##
## data:  x by post_stim$StimType (Sham, TI 1:3)
## Z = -1.0625, p-value = 0.288
## alternative hypothesis: true mu is not equal to 0
##
##
## $adv_pain
##
##  Asymptotic Wilcoxon-Mann-Whitney Test
```

```
##
## data:  x by post_stim$StimType (Sham, TI 1:3)
## Z = 0.47077, p-value = 0.6378
## alternative hypothesis: true mu is not equal to 0
##
##
## $adv_burning
##
##  Asymptotic Wilcoxon-Mann-Whitney Test
##
## data:  x by post_stim$StimType (Sham, TI 1:3)
## Z = -1, p-value = 0.3173
## alternative hypothesis: true mu is not equal to 0
##
##
## $`adv_burning-heat`
##
##  Asymptotic Wilcoxon-Mann-Whitney Test
##
## data:  x by post_stim$StimType (Sham, TI 1:3)
## Z = -0.59197, p-value = 0.5539
## alternative hypothesis: true mu is not equal to 0
##
##
## $adv_metallictaste
##
##  Asymptotic Wilcoxon-Mann-Whitney Test
##
## data:  x by post_stim$StimType (Sham, TI 1:3)
## Z = NaN, p-value = NA
## alternative hypothesis: true mu is not equal to 0
##
##
## $adv_fatigue
##
##  Asymptotic Wilcoxon-Mann-Whitney Test
##
## data:  x by post_stim$StimType (Sham, TI 1:3)
## Z = -0.11192, p-value = 0.9109
## alternative hypothesis: true mu is not equal to 0
##
##
## $`adv_fatigue-sleepiness`
##
##  Asymptotic Wilcoxon-Mann-Whitney Test
##
## data:  x by post_stim$StimType (Sham, TI 1:3)
## Z = -0.47258, p-value = 0.6365
## alternative hypothesis: true mu is not equal to 0
##
##
## $adv_mood
##
##  Asymptotic Wilcoxon-Mann-Whitney Test
```

```
##
## data:  x by post_stim$StimType (Sham, TI 1:3)
## Z = NaN, p-value = NA
## alternative hypothesis: true mu is not equal to 0
##
##
## $adv_anxiety
##
##  Asymptotic Wilcoxon-Mann-Whitney Test
##
## data:  x by post_stim$StimType (Sham, TI 1:3)
## Z = -1, p-value = 0.3173
## alternative hypothesis: true mu is not equal to 0
##
##
## $adv_discomfort
##
##  Asymptotic Wilcoxon-Mann-Whitney Test
##
## data:  x by post_stim$StimType (Sham, TI 1:3)
## Z = -1.4318, p-value = 0.1522
## alternative hypothesis: true mu is not equal to 0
##
##
## $adv_unpleasant
##
##  Asymptotic Wilcoxon-Mann-Whitney Test
##
## data:  x by post_stim$StimType (Sham, TI 1:3)
## Z = NaN, p-value = NA
## alternative hypothesis: true mu is not equal to 0
##
##
## $adv_dizziness
##
##  Asymptotic Wilcoxon-Mann-Whitney Test
##
## data:  x by post_stim$StimType (Sham, TI 1:3)
## Z = NaN, p-value = NA
## alternative hypothesis: true mu is not equal to 0
##
##
## $adv_nausea
##
##  Asymptotic Wilcoxon-Mann-Whitney Test
##
## data:  x by post_stim$StimType (Sham, TI 1:3)
## Z = -1, p-value = 0.3173
## alternative hypothesis: true mu is not equal to 0
##
##
## $adv_visual
##
##  Asymptotic Wilcoxon-Mann-Whitney Test
```

```
##
## data:  x by post_stim$StimType (Sham, TI 1:3)
## Z = NaN, p-value = NA
## alternative hypothesis: true mu is not equal to 0
```

```r
data_long <- gather(post_stim, side_effect, rating, adv_itching:adv_visual, factor_key=TRUE)


### Means to report
stat <-  data_long  %>%
  group_by(StimType, side_effect) %>%
  summarise_at(vars(rating), list(mean = mean, std = sd, min = min, max = max))

pander(stat)
```

StimType	side_effect	mean	std	min	max
Sham	adv_itching	1	0	1	1
Sham	adv_tingling	1.476	0.6016	1	3
Sham	adv_skinredness	1	0	1	1
Sham	adv_headache	1.048	0.2182	1	2
Sham	adv_pain	1.143	0.3586	1	2
Sham	adv_burning	1	0	1	1
Sham	adv_burning-heat	1.048	0.2182	1	2
Sham	adv_metallictaste	1	0	1	1
Sham	adv_fatigue	1.619	0.8047	1	3
Sham	adv_fatigue-sleepiness	1.571	0.8106	1	3
Sham	adv_mood	1	0	1	1
Sham	adv_anxiety	1	0	1	1
Sham	adv_discomfort	1	0	1	1
Sham	adv_unpleasant	1	0	1	1
Sham	adv_dizziness	1	0	1	1
Sham	adv_nausea	1	0	1	1
Sham	adv_visual	1	0	1	1
TI 1:3	adv_itching	1.238	0.4364	1	2
TI 1:3	adv_tingling	1.476	0.6016	1	3
TI 1:3	adv_skinredness	1	0	1	1
TI 1:3	adv_headache	1.19	0.5118	1	3
TI 1:3	adv_pain	1.095	0.3008	1	2
TI 1:3	adv_burning	1.048	0.2182	1	2
TI 1:3	adv_burning-heat	1.095	0.3008	1	2
TI 1:3	adv_metallictaste	1	0	1	1
TI 1:3	adv_fatigue	1.619	0.74	1	3
TI 1:3	adv_fatigue-sleepiness	1.714	0.9024	1	3
TI 1:3	adv_mood	1	0	1	1
TI 1:3	adv_anxiety	1.048	0.2182	1	2
TI 1:3	adv_discomfort	1.095	0.3008	1	2
TI 1:3	adv_unpleasant	NA	NA	NA	NA
TI 1:3	adv_dizziness	1	0	1	1
TI 1:3	adv_nausea	1.048	0.2182	1	2
TI 1:3	adv_visual	1	0	1	1